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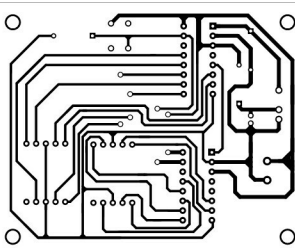


Fig. 5: Actual-size PCB pattern of the receiver circuit

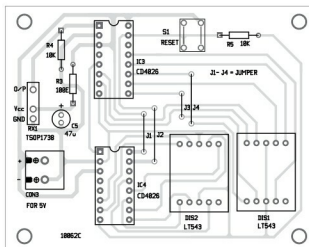


Fig. 6: Component layout for the PCB in Fig. 5

The formula for the frequency is given by $f = 1.44 / (R1 + 2VR1)C3$. You can vary VR1 to get the exact frequency. The IR LED connected to output pin (pin 3) of IC2 transmits these frequency pulses (IR beams), which are received by the TSOP1738 IR receiver (RX1).

When you interrupt the frequency pulses, immediately the IR receiver (TSOP1738) generates a clock pulse, which is fed to decade counter IC3 at its pin 1 (CLK).

To count each clock pulse the output of IC3 is fed to the 7-segment display, which displays equivalent clock pulse in digital format. After leading edge of every tenth clock pulse it simply resets the count to

zero position. The carry output from pin 5 of IC3 is fed to pin 1 (CLK) of IC4. This increases the number displayed in ten's display by one.

This way the process continues to display up to 99. After 99 the counter simply resets to 00. To reset the display at any time press reset switch S1. To increase the circuit's capacity to count up to 999, simply add another CD4026 IC and a 7-segment display in a similar fashion.

Construction and testing

An actual-size, single-side PCB pattern for the transmitter section is shown in Fig. 3 and its component layout in Fig. 4. Similarly, an

actual-size, single-side PCB of the receiver section is shown in Fig. 5 and its component layout in Fig. 6.

Install the transmitter and receiver units at the gate such that the IR beam of IR LED1 falls directly on the IR receiver RX1. Whenever a person crosses the gate IR beam is interrupted. Each interruption generates a clock pulse in the receiver unit, which is registered as a count and is displayed on the 7-segment displays. **EFY**



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